

M. BALANDARI TOROGHI

m.balandari.toroghi@gmail.com - [linkedin.com/in/mbalandari](https://www.linkedin.com/in/mbalandari) - <https://mbalandari.github.io>

Mining Engineer with extensive experience across underground mining, rock mechanics, geotechnical engineering, mineral exploration, and tunneling. Experienced in rock-mass characterization, laboratory testing, ground-stability assessment, numerical modeling (RS2/RS3), and underground excavation evaluation. Combining engineering expertise with Python, data analysis, and engineering software development to support mining digitalization and data-driven decision making.

EXPERIENCE & EDUCATION

2022– PRESENT

MINING DIGITALIZATION & DATA ANALYTICS | PROFESSIONAL DEVELOPMENT

- Applied Python to mining and geotechnical datasets for automated data processing, statistical analysis, and technical reporting.
- Developed computational workflows for interpretation of rock-mechanics laboratory data, including UCS, BTS, triaxial, and durability testing.
- Studied data analytics and machine-learning applications in underground mining, ground control, and predictive maintenance.
- Developed three Python-based engineering applications for rock-mass classification and laboratory test data analysis.

PROJECTS - ENGINEERING SOFTWARE & DATA ANALYTICS

- **Rock Mass Classification Tool** - Python/PySide6 application implementing **RMR, Q-System and GSI**, with visualization, preliminary support assessment, and automated engineering reports.
- **Triaxial Test Analyzer** - Python/PySide6 application for **triaxial rock-mechanics analysis**, including peak strength, principal stresses, Mohr circles, and Mohr-Coulomb failure-envelope fitting.
- **UCS Analyzer** - Python/PySide6 application for **UCS laboratory data analysis**, including stress-strain processing, UCS, Young's modulus, visualization, and automated reporting.

2017 – 2021

RESEARCH ASSISTANT, ISTANBUL TECHNICAL UNIVERSITY

- Planned, supervised, and interpreted rock-mechanics laboratory testing programs including UCS, BTS, triaxial compression, direct shear, slake durability, and porosity testing.
- Performed rock-mass characterization and geotechnical analyses supporting underground excavation stability, support design, mining-method selection, and tunnel assessment.
- Conducted FEM-based numerical modeling of underground tunnels using RS2/RS3 and compared results with analytical methods to evaluate and optimize tunnel designs.
- Prepared technical reports and engineering recommendations for mining and underground construction projects.

2015 – 2016

PROFESSIONAL DEVELOPMENT IN MINING ENGINEERING

- Advanced study of underground mining software, rock mechanics, numerical/analytical ground-stability methods, international mining standards, and ground-support principles.

2012 – 2014

SITE GEOTECHNICAL ENGINEER, STRATUS INTERNATIONAL CONTRACTING COMPANY

- Supported tunneling operations at the Azad Dam project through geological and geotechnical mapping of underground excavations.
- Performed rock-mass classification using **RMR, Q-System, and GSI** and assisted with ground-support and excavation-stability assessment.
- Monitored geotechnical conditions and excavation performance during construction, coordinating with construction and engineering teams to support safe tunnel development.

2009 – 2011

MS MINING ENGINEERING, AMIRKABIR UNIVERSITY OF TECHNOLOGY (TEHRAN POLYTECHNIC)

- Master Thesis Title: Technical Analysis of Cut-off Grouting Requirement According to Multi Attribute Decision Making (Case Study: Aghchai and Talvar Dam Sites)
- Supervisor: Assoc. Professor Dr. Homayoon Katibeh, Advisor: Dr. Iraj Golabatunchi
- Grade A

2006 – 2008

FIELD MINING ENGINEER, BEH AVARAN MINING COMPANY

- Participated in mineral exploration campaigns involving geological mapping, prospecting, geochemical sampling, target generation, and prospectivity assessment.
- Prepared geological maps, sampling databases, geochemical anomaly maps, and technical reports.

2005 – 2006

ENGINEERING INTERN, TECHNICAL SOIL & ROCK MECHANICS LABORATORY

- Undertook an internship at the central rock mechanics laboratory of the province, conducting standard soil, rock, and concrete tests such as uniaxial and triaxial compression tests on rock cores, soil gradation analysis, and concrete mix design.

2000 – 2005

BS MINING ENGINEERING - EXPLORATION, UNIVERSITY OF BIRJAND

- Focused on mineral exploration operations including geochemical field exploration work

TECHNICAL SKILLS

- Rock Mechanics
- Geological Mapping
- Mine Geotechnics
- Python
- PySide6
- Tunnel Engineering
- Rock Mass Classification (RMR, Q, GSI)
- Data Analysis
- Pandas
- Underground Mining
- Ground Stability Analysis
- Numerical Modeling
- Scientific Visualization
- Engineering Software Development

LANGUAGES

- English – Fluent, Persian – Native, French – Basic, Turkish – Basic

LICENSES & CERTIFICATIONS

- ✓ GIS Basics & ArcGIS Pro Basics - Issuing organization: Esri - Credential ID 05162021
- ✓ Data Analysis with Python (IBM Badge) - Issued by Coursera, Authorized by IBM, [See credential](#)
- ✓ Python for Everybody Specialization - Issued by Coursera, Authorized by University of Michigan, [See credential](#)
- ✓ TOEFL iBT (94 Total), GRE (Quantitative Reasoning (160), Verbal Reasoning (147))